

Passenger No-Show Analysis and Prediction

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Dataset: AirLincolnData2

Task 1: Data Auditing, Metadata Governance & Feature Hygiene

1.1 Data Audit and Baseline No-Show Rate

AirLincolnData2 contains 10,000 passenger observations.

The response variable No_Show has a minimum of 0, maximum of 1, mean of 0.0998, and 998 no-show observations out of 10,000 records. This corresponds to a baseline no-show rate of 9.98% (998/10,000). The 9.98% baseline event rate provides an important benchmark for evaluating model performance and for selecting a prediction cutoff that is appropriate for the relatively rare no-show event.

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Model Information

Dimensions

Response Profile

Iteration History

Convergence

Fit Statistics

Parameter Estimates

Ty

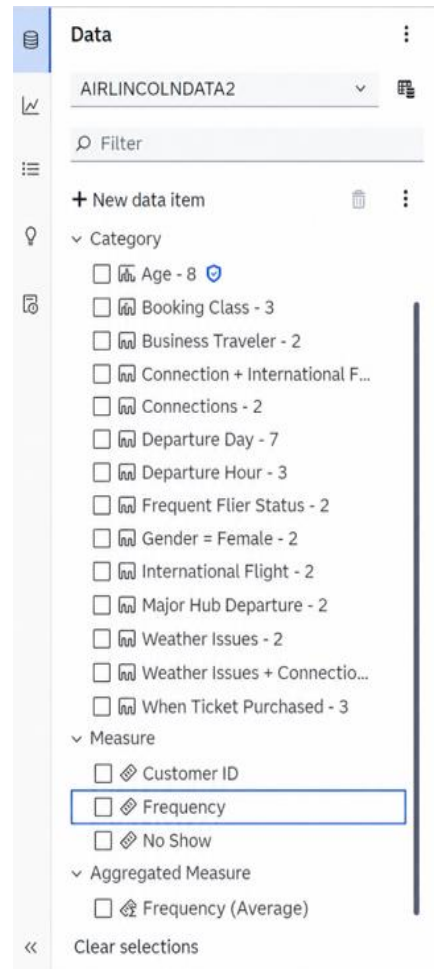
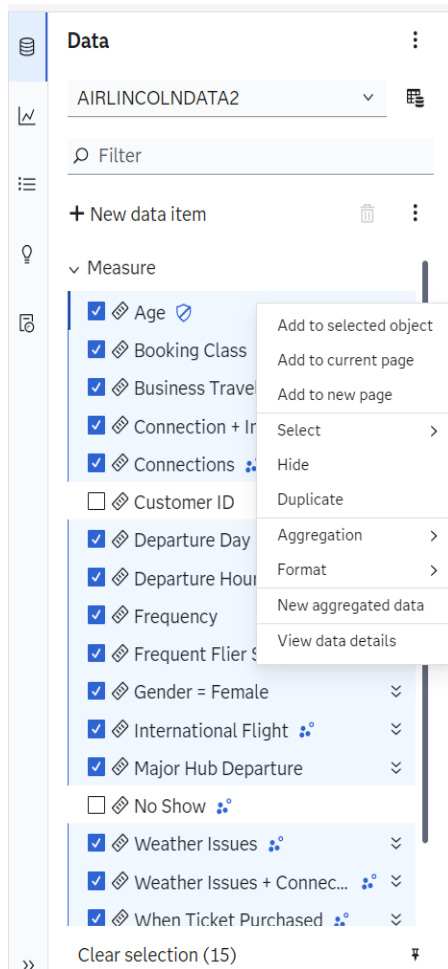
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Ordered Value	Count	No show category
1	998	No show
2	9002	Show

1.2 Metadata Reclassification

All 15 required predictor variables were treated as categorical variables for modeling. The predictors contain coded discrete groups rather than measurements for which a linear distance between codes should automatically be assumed.

Reclassifying the coded predictors as categories prevents erroneous linear assumptions. For example, Age groups coded 1–8 and Booking Class coded 1–3 represent discrete groups; the numerical labels do not necessarily imply equal-interval relationships. Treating them as categories allows the models to estimate differences between groups without imposing an artificial linear structure.



1.3 Feature Exclusion & Noise Reduction

Raw Customer_ID and other uninformative or redundant raw fields were excluded from the predictive effects. Customer_ID is an arbitrary record identifier rather than a repeatable passenger or flight characteristic. Leaving such an identifier active can introduce noise and may allow a tree-based model to create record-specific patterns that do not generalize to future observations.

The data-preparation evidence shows raw fields outside the required predictor set while the modeling predictors remain available for analysis.

Task 2: Partitioning & Multi-Model Estimation

2.1 Honest Assessment Setup

A Train_Validation_Partition variable was created using stratified sampling. The partition uses 60% of the observations for Training and 40% for Validation, with the random seed set to 12345. Stratification was applied across No_Show, Booking_Class, Business_Traveler, and Weather_Issues.

Stratified sampling helps maintain a similar distribution of the target and important predictor groups across the training and validation samples. This reduces the risk that an unbalanced random split produces an unusually easy or difficult validation sample, which could lead to an overly optimistic assessment of model performance. Evaluating the models on a separate validation sample therefore provides a more honest estimate of how well the models generalize to unseen observations

Edit Partition

Name: *

Train_Validation_Partition

Based on: ⓘ

☐ Data item ☒ Sampling

Sampling method:

Stratified sampling ▾

Stratify by (up to 4 items):

- ☒ No show category
- ☒ Booking Class
- ☒ Business Traveler
- ☒ Weather Issues



+ Add

Number of partitions:

2 ▾

Training: * 60 %

OK

Cancel

+ Add

Number of partitions:

2

Training: *

60

%

Validation: 40 %

☒ Random number seed

Random seed: *

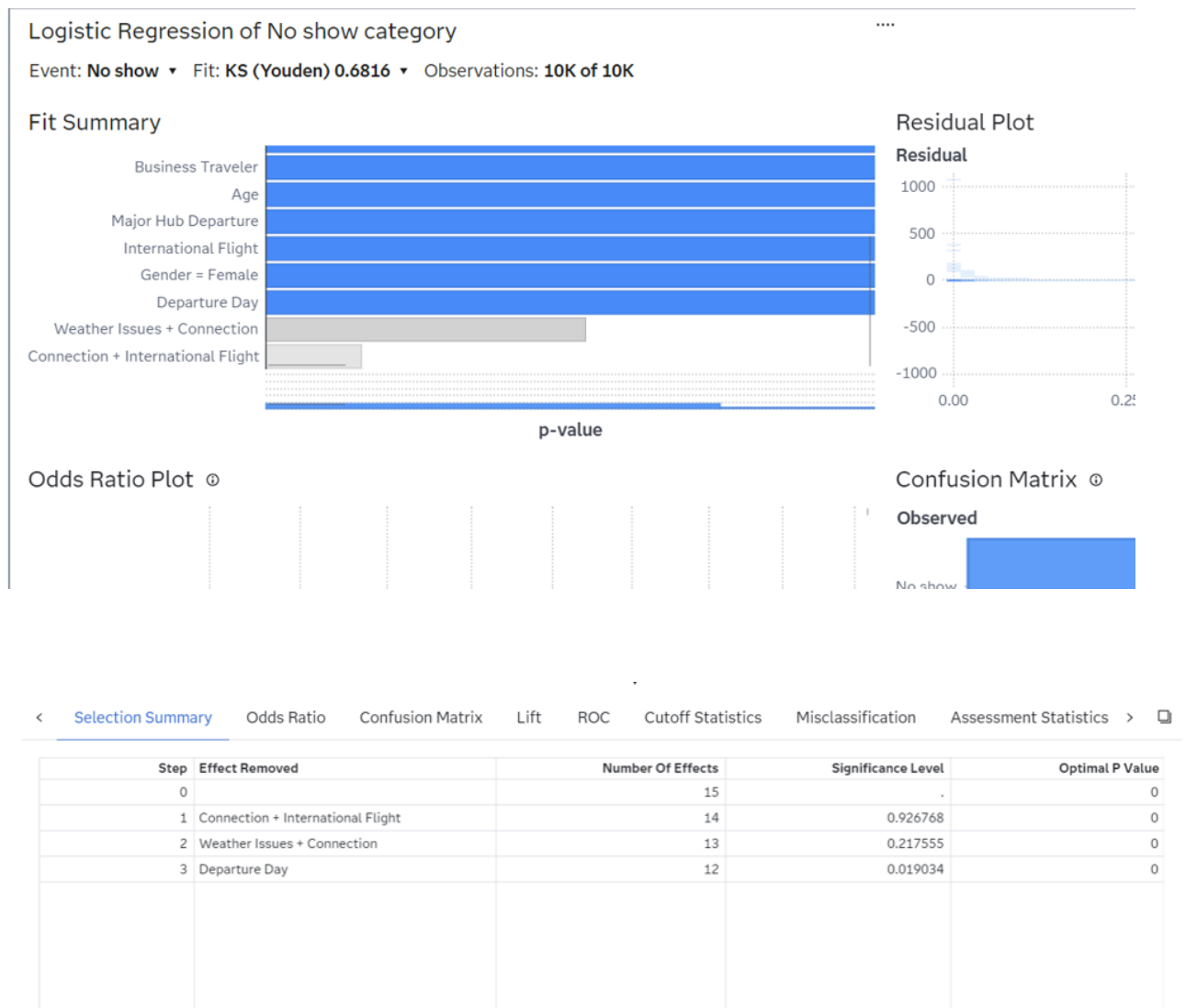
12,345

A unique ID is required to create deterministic partitions.

2.2 Logistic Regression

A Logistic Regression model was estimated using the No show category as the response and the 15 predictor variables as classification effects. The model used the Train_Validation_Partition variable to evaluate performance on the validation sample. Fast Backward selection was used to reduce unnecessary predictors and identify a more focused model specification.

The Logistic Regression model achieved a Validation KS (Youden) of 0.6581. The model's variable-selection output indicates that the selected predictors were associated with the response at different levels of statistical significance.

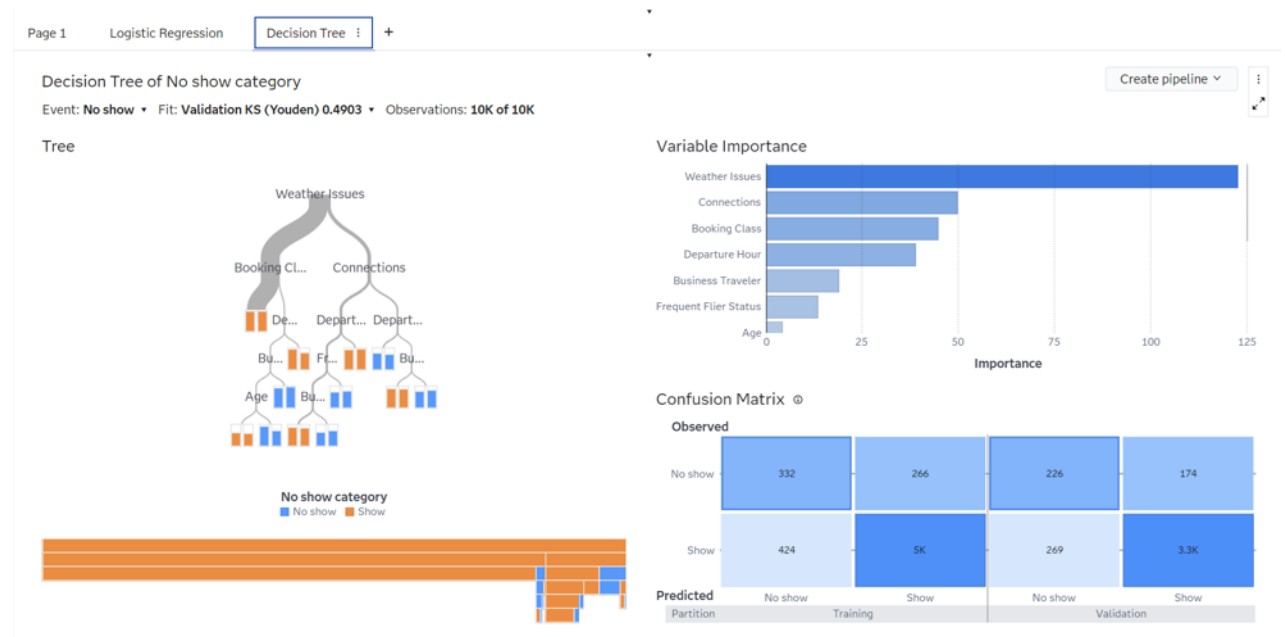


2.3 Decision Tree

A Decision Tree was estimated using the same response, event definition, predictors, and Train_Validation_Partition. The minimum Leaf Size was set to 10.

The minimum leaf size of 10 prevents terminal nodes from becoming extremely small and reduces the risk of fitting highly specific patterns or random noise in the training sample. At the same time, the restriction preserves sufficient flexibility for the tree to identify meaningful relationships and supports better generalization to the validation sample.

The Decision Tree achieved a Validation KS (Youden) of 0.4903. Its variable-importance display ranked Weather Issues, Connections, and Booking Class among the most important predictors.



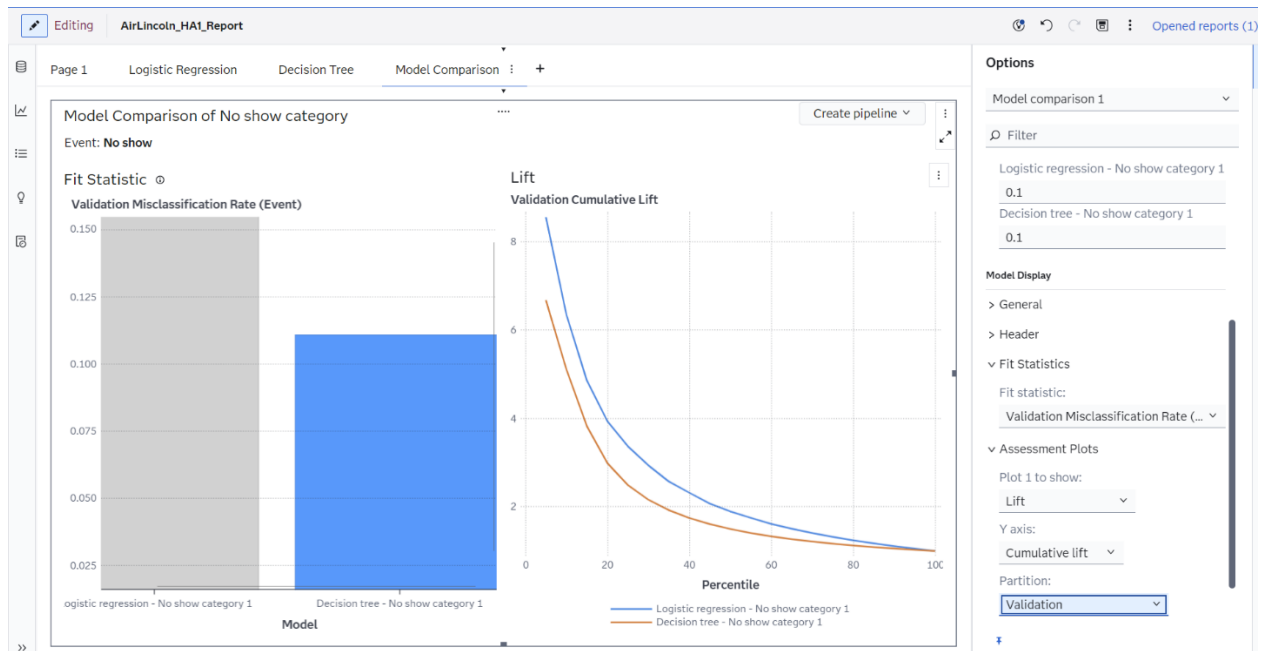
2.4 Hyperparameter Rationale

The Decision Tree's minimum Leaf Size = 10 was selected to control model complexity. Without a minimum leaf-size constraint, the tree could continue splitting observations into very small terminal nodes, allowing the model to capture noise or unusual patterns specific to the training data.

Requiring at least 10 observations per terminal node provides a balance between model flexibility and overfitting control. The resulting model can still identify nonlinear relationships and interactions while reducing the likelihood that its predictions depend on very small groups that may not generalize to the validation sample.

Task 3: Champion Selection & Goodness-of-Fit Benchmarking

Both models were compared on the validation sample using Prediction Cutoff = 0.10. For cumulative lift, Percentile = 20%. SAS labels the available event-error metric in this version as "Validation Misclassification Rate (Event)."



Validation Metric	Logistic Regression	Decision Tree
Misclassification Rate (Event)	0.1588	0.1108
KS (Youden)	0.6581	0.4903
False Positive Rate	0.1547	0.0747
Cumulative Lift 20%	3.9375	2.9899

For the selected 0.10 cutoff, the Decision Tree is designated as the champion model based on its lower validation misclassification rate and lower false positive rate. However, Logistic Regression demonstrates stronger KS and cumulative lift, so the choice reflects the operational priority placed on minimizing overall classification and false-positive errors rather than superior performance on every metric.

Strategic Business Implications & Governance

Q1. Risk Asymmetry & Overbooking Economics:

A false positive occurs when the model predicts a passenger will not show but the passenger actually arrives. If the prediction supports overbooking, excessive false positives can contribute to oversold flights, denied boarding, compensation, rebooking costs, and customer dissatisfaction.

A false negative occurs when the model predicts that a passenger will show but the passenger actually no-shows. This can leave an empty seat and reduce realized revenue. The two errors therefore have different business consequences.

The 0.10 prediction cutoff makes the models more willing to flag potential no-shows than a conventional 0.50 cutoff. This is relevant because no-shows are relatively rare. However, a lower threshold can also increase the number of passengers flagged as potential no-shows. The cutoff should ultimately be calibrated using AirLincoln's actual financial cost of an empty seat versus the cost of an incorrect no-show prediction.

Q2. Variable Importance & Targeted Interventions:

The Decision Tree variable-importance results identify Weather Issues, Connections, and Booking Class as the three strongest predictive drivers. These are predictive associations and should not be interpreted as proof of causation.

Intervention 1: Risk-Based Pre-Departure Confirmation

AirLincoln could use predicted no-show risk to identify passengers who may benefit from targeted confirmation or itinerary reminders, especially when weather disruption or connecting itineraries are involved. The message could request confirmation of travel intent and surface rebooking options when disruption risk is elevated.

Intervention 2 : Flight-Level Risk Planning

Passenger-level risk scores could be aggregated at the flight level to inform controlled overbooking and standby planning. The scores should be combined with route, capacity, disruption, and compensation constraints. Human revenue-management rules should remain in place for decisions that affect boarding or customer treatment.

Q3. AI Copilot Critique & Governance Review:

The SAS-generated Copilot summary is useful as a first-pass explanation because it quickly identifies key validation metrics and translates model output into plain language. However, the AI-generated interpretation should be checked against the underlying SAS output.

Statistical significance does not automatically imply operational importance. Model quality also depends on validation evidence, class imbalance, the selected cutoff, false-positive and false-negative costs, and performance on unseen data.

A responsible workflow requires a human analyst to verify AI-generated interpretations against the SAS output, document the validation sample and cutoff,

evaluate stability across relevant passenger and route segments, and monitor model drift after deployment. Copilot should support interpretation rather than act as the final authority for overbooking or other customer-impact decisions.

 Generated Model Summary

Logistic Regression of No show category

The logistic regression model appears to be highly performant, as indicated by the strong statistical significance of all included effects, each with a p-value well below the 0.05 threshold. Effects such as Weather Issues, Connections, When Ticket Purchased, Booking Class, Departure Hour, Frequent Flier Status, Business Traveler, Major Hub Departure, Age, International Flight, and Gender (Female) are all significant contributors to the model. The odds ratios suggest that these variables may have substantial impacts on the predicted outcome, and the lift data indicates the model is able to effectively distinguish between high and low risk groups. Overall, the model may provide reliable predictions, but results should be interpreted with some caution as real-world performance can vary.

AI-generated content. Evaluate accuracy and suitability before use.

Close

SAS Viya Copilot

Review the Model Comparison results for AirLincoln's passenger no-show prediction analysis. Summarize the results for an airline manager, including the Validation Misclassification Rate, Validation KS (Youden), Validation False Positive Rate, and Cumulative Lift at 20% for both Logistic Regression and Decision Tree. Explain the business implication of the results, including the trade-off between false positives and false negatives and the use of a 0.10 prediction cutoff. Identity any limitations, assumptions, or areas where additional human judgement is needed. Do not make claims of causation unless supported by the analysis.

 Here's a summary of the Model Comparison results for AirLincoln's passenger no-show prediction analysis, focusing on the validation partition and using a 0.10 prediction cutoff for both models:

Key Metrics (Validation Partition, Cutoff = 0.10)		
Metric	Logistic Regression	Decision Tree (Champion)
Validation Misclassification Rate	0.15875	0.11075
Validation KS (Youden)	0.65806	0.49028
Validation False Positive Rate	0.15472	0.07472
Cumulative Lift at 20%	3.94 	2.99

Interpretation for Airline Management

- **Model Performance:**

The Decision Tree model outperformed Logistic Regression on the validation data, with a lower misclassification rate (11.1% vs. 15.9%). This means the Decision Tree made fewer overall errors in predicting passenger no-shows.

- **Validation KS (Youden):**

The KS statistic, which measures the model's ability to separate no-shows from shows, is higher for Logistic Regression (0.66) than for the Decision Tree (0.49). This suggests Logistic Regression has slightly better discrimination, but the Decision Tree's lower misclassification rate makes it preferable for operational use.

- **False Positive Rate:**

The Decision Tree's false positive rate (7.5%) is about half that of Logistic Regression (15.5%). A false positive here means predicting a passenger will no-show when they actually show up. Lowering this rate reduces unnecessary interventions (e.g., overbooking or customer outreach).

- **Cumulative Lift at 20%:**

Cumulative Lift at 20% indicates how much better the model is at identifying no-shows in the top 20% of predicted risk compared to random selection. Logistic Regression achieves a higher lift (3.94) than the Decision Tree (2.99), meaning it is more effective at concentrating true no-shows in the highest-risk group.

Business Implications and Trade-offs

- **False Positives vs. False Negatives:**

- **False Positives (FP):** Predicting a no-show when the passenger actually shows up can lead to overbooking or unnecessary customer contact. The Decision Tree's lower FP rate reduces these risks.
- **False Negatives (FN):** Predicting a show when the passenger actually no-shows can result in lost revenue or empty seats. Both models have a trade-off between FP and FN, controlled by the cutoff.

- **Cutoff Value (0.10):**

Using a 0.10 cutoff means passengers with a predicted no-show probability above 10% are flagged as likely no-shows. This is a relatively low threshold, favoring sensitivity (catching more true no-shows) at the expense of more false positives. The cutoff can be adjusted based on business priorities (e.g., minimizing denied boarding vs. maximizing seat utilization).

- **Model Selection:**

The Decision Tree was selected as the champion model due to its lower misclassification rate at the chosen cutoff, even though Logistic Regression had a higher KS and lift. This reflects a business preference for fewer total errors and lower operational risk from false positives.

Limitations, Assumptions, and Need for Human Judgement

- **Assumptions:**
 - The models assume that historical patterns of no-shows will continue in the future.
 - The cutoff value (0.10) is chosen based on current business objectives and may not be optimal for all scenarios.
 - Model inputs (features) are assumed to be accurate and consistently available.
- **Limitations:**
 - The models do not account for sudden changes in passenger behavior (e.g., due to pandemics, policy changes).
 - The lift and KS statistics are based on the validation set and may not generalize perfectly to future data.
 - The analysis does not establish causation—important factors (e.g., weather, connections) are associated with no-shows but are not necessarily the cause.
- **Areas for Human Judgement:**
 - Adjusting the cutoff to balance business risk (e.g., cost of empty seats vs. customer dissatisfaction).
 - Interpreting model predictions in the context of operational realities (e.g., flight capacity, overbooking policies).
 - Monitoring for model drift and retraining as needed.

Summary

- The Decision Tree model is recommended for operational use due to its lower error rate and false positive rate at the 0.10 cutoff.
- Logistic Regression offers better lift and separation, which may be useful for targeted interventions.
- The choice of cutoff and model should be revisited as business needs evolve, and results should be monitored for consistency over time.
- Human oversight is essential for interpreting results, setting thresholds, and responding to changes in passenger behavior.

If you need a visual summary or want to explore how changing the cutoff affects these metrics, let me know!



Mandatory AI Usage Disclosure Paragraph:

I used ChatGPT and SAS Viya Copilot as a learning and writing support tool during this assignment.

I used ChatGPT to organize my written analysis and review my interpretation of SAS model outputs to make sure I meet all the deliverables. SAS Viya Copilot was used to generate a first-pass summary of the Model Comparison results and their business implications. I did not upload the AirLincoln course dataset or other course data to ChatGPT. The dataset, model estimation, partitioning, validation metrics, and screenshots were generated and verified directly in SAS Visual Analytics.

The main prompts I used with ChatGPT included: The SAS Viya Copilot portion, I used the following prompt in SAS: “Review the Model Comparison results for AirLincoln’s passenger no-show prediction analysis. Summarize the results for an airline manager, including the Validation Misclassification Rate, Validation KS (Youden), Validation False Positive Rate, and Cumulative Lift at 20% for both Logistic Regression and Decision Tree. Explain the business implications of the results, including the trade-off between false positives and false negatives and the use of a 0.10 prediction cutoff. Identify any limitations, assumptions, or areas where additional human judgment is needed. Do not make claims of causation unless supported by the analysis.”, “Check my assignment against the professor's requirements and identify any missing deliverables.”

Verifying and refining the AI outputs taught me that AI-generated explanations should not be accepted without checking them against the actual statistical output.

I learned that different evaluation metrics can favor different models: Logistic Regression had higher validation KS and cumulative lift, while the Decision Tree had lower misclassification and false positive rates at the selected cutoff. I also learned that model outputs describe predictive associations rather than necessarily establishing causation, and that business decisions such as overbooking require human judgment regarding the costs of false positives, false negatives, customer impact, and operational constraints.